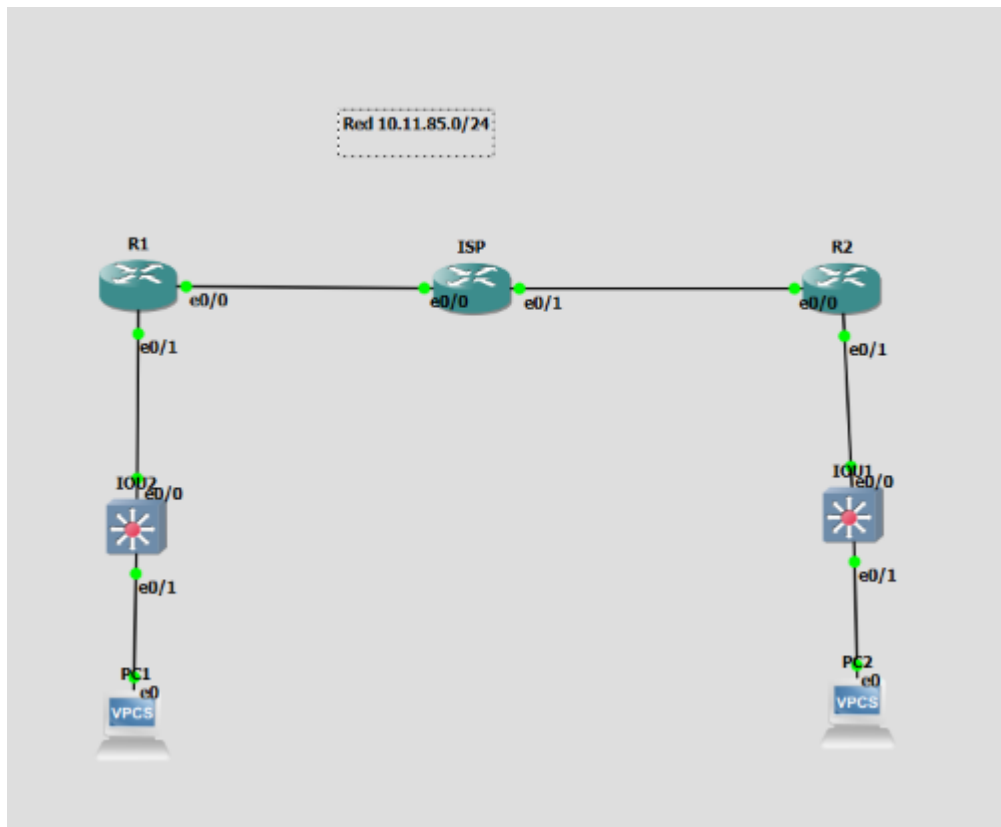


Objetivo

Implementar una VPN Site-to-Site usando **GRE over IPSec con IKEv1**. Se combinan dos tecnologías: **GRE** crea un túnel lógico que soporta multicast, broadcast y cualquier protocolo de capa 3 (permitiendo enrutamiento dinámico como OSPF/EIGRP); e **IPSec IKEv1 en modo transport** cifra el tráfico GRE entre los peers WAN, garantizando confidencialidad e integridad. Esta combinación es la base histórica de DMVPN.

Topología



Direccionamiento IP

Dispositivo	Interfaz	Dirección IP	Descripción
R1	e0/0	10.11.85.65/26	WAN → ISP

R1	e0/1	10.11.85.1/26	LAN → SW1
R1	Tunnel0	10.10.10.1/30	GRE hacia R2
ISP	e0/0	10.11.85.66/26	Hacia R1
ISP	e0/1	10.11.85.129/26	Hacia R2
R2	e0/0	10.11.85.130/26	WAN → ISP
R2	e0/1	10.11.85.193/26	LAN → SW2
R2	Tunnel0	10.10.10.2/30	GRE hacia R1
PC1	e0	10.11.85.2/26	GW: 10.11.85.1
PC2	e0	10.11.85.194/26	GW: 10.11.85.193

Parámetros IPsec IKEv1

Fase 1 — IKE / ISAKMP

Parámetro	Valor
Política	10
Cifrado	AES-256
Hash	SHA-256
Autenticación	Pre-Shared Key
Grupo DH	14 (2048-bit)
Lifetime	86400 segundos
Clave PSK	ITLA2024

Fase 2 — IPsec (diferencia clave: modo TRANSPORT)

Parámetro	Valor
Transform Set	TS-GRE

Protocolo	ESP
Cifrado	AES-256
Integridad	HMAC-SHA-256
Modo	TRANSPORT (GRE ya encapsula, no se necesita tunnel)
ACL tráfico	permit gre host 10.11.85.65 host 10.11.85.130
Crypto Map	CM-GRE (en Ethernet0/0)

Tunnel0 — GRE

Parámetro	R1	R2
IP Tunnel	10.10.10.1/30	10.10.10.2/30
Tunnel source	Ethernet0/0	Ethernet0/0
Tunnel destination	10.11.85.130	10.11.85.65
Tunnel mode	gre ip	gre ip

Funcionamiento — GRE over IPSec

El flujo de datos tiene dos etapas de procesamiento en R1:

ETAPA 1 — GRE encapsula:

PC1 envía: src 10.11.85.2 dst 10.11.85.194

R1: ruta 10.11.85.192/26 → via Tunnel0 (GRE)

GRE agrega header: src 10.11.85.65 dst 10.11.85.130 proto 47

ETAPA 2 — IPSec cifra el paquete GRE: ACL-GRE-VPN matchea: permit gre host

10.11.85.65 host 10.11.85.130 ESP (modo transport) cifra el payload GRE con AES-256

Sale por e0/0 → ISP → R2

R2: IPSec descifra → GRE desencapsula → entrega a PC2

■ **Modo transport vs tunnel:** GRE ya provee el IP outer (10.11.85.65→.130). Usar modo tunnel agregaría un segundo IP outer innecesario. Transport solo cifra el payload, evitando overhead extra.

Scripts de Configuración

ISP

configure terminal

hostname ISP

interface Ethernet0/0

description Hacia R1

ip address 10.11.85.66 255.255.255.192

no shutdown

interface Ethernet0/1

description Hacia R2

ip address 10.11.85.129 255.255.255.192

no shutdown

end

R1

enable

configure terminal

hostname R1

interface Ethernet0/0

description WAN hacia ISP

ip address 10.11.85.65 255.255.255.192

no shutdown

interface Ethernet0/1

description LAN hacia SW1-PC1

ip address 10.11.85.1 255.255.255.192

no shutdown

ip route 0.0.0.0 0.0.0.0 10.11.85.66

interface Tunnel0

description GRE Tunnel hacia R2

ip address 10.10.10.1 255.255.255.252

tunnel source Ethernet0/0

```
tunnel destination 10.11.85.130
tunnel mode gre ip
no shutdown
```

```
ip route 10.11.85.192 255.255.255.192 Tunnel0
```

```
crypto isakmp policy 10
  encr aes 256
  hash sha256
  authentication pre-share
  group 14
  lifetime 86400
```

```
crypto isakmp key ITLA2024 address 10.11.85.130
```

```
crypto ipsec transform-set TS-GRE esp-aes 256 esp-sha256-hmac
mode transport
ip access-list extended ACL-GRE-VPN
permit gre host 10.11.85.65 host 10.11.85.130
```

```
crypto map CM-GRE 10 ipsec-isakmp
set peer 10.11.85.130
set transform-set TS-GRE
match address ACL-GRE-VPN
```

```
interface Ethernet0/0
crypto map CM-GRE
```

```
end
```

R2

```
enable
configure terminal
```

```
hostname R2
```

```
interface Ethernet0/0
description WAN hacia ISP
ip address 10.11.85.130 255.255.255.192
no shutdown
```

```
interface Ethernet0/1
description LAN hacia SW2-PC2
ip address 10.11.85.193 255.255.255.192
no shutdown
```

```
ip route 0.0.0.0 0.0.0.0 10.11.85.129
```

```
interface Tunnel0
description GRE Tunnel hacia R1
ip address 10.10.10.2 255.255.255.252
tunnel source Ethernet0/0
tunnel destination 10.11.85.65
tunnel mode gre ip
no shutdown
ip route 10.11.85.0 255.255.255.192 Tunnel0
```

```
crypto isakmp policy 10
encr aes 256
hash sha256
authentication pre-share
group 14
lifetime 86400
```

```
crypto isakmp key ITLA2024 address 10.11.85.65
crypto ipsec transform-set TS-GRE esp-aes 256 esp-sha256-hmac
mode transport
ip access-list extended ACL-GRE-VPN
permit gre host 10.11.85.130 host 10.11.85.65
```

```
crypto map CM-GRE 10 ipsec-isakmp
set peer 10.11.85.65
set transform-set TS-GRE
match address ACL-GRE-VPN
```

```
interface Ethernet0/0
crypto map CM-GRE
```

```
end
```

VPCS

```
! PC1
ip 10.11.85.2 255.255.255.192 10.11.85.1 save
```

```
! PC2
```

```
ip 10.11.85.194 255.255.255.192 10.11.85.193 save
```

Verificación

```
! Tunnel0 UP/UP, tunnel mode GRE/IP show
interface Tunnel0
```

```
R2#sh int tunnel 0
Tunnel0 is up, line protocol is up
  Hardware is Tunnel
  Description: GRE Tunnel hacia R1
  Internet address is 10.10.10.2/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 10.11.85.130 (Ethernet0/0), destination 10.11.85.65
  Tunnel Subblocks:
    src-track:
      Tunnel0 source tracking subblock associated with Ethernet0/0
      Set of tunnels with source Ethernet0/0, 1 member (includes iterators), on interface <OK>
  Tunnel protocol/transport GRE/IP
  Key disabled, sequencing disabled
  Checksumming of packets disabled
  Tunnel TTL 255, Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input 00:02:12, output 00:02:12, output hang never
  Last clearing of "show interface" counters 00:26:23
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    44 packets input, 4832 bytes, 0 no buffer
    Received 0 broadcasts (0 IP multicasts)
    0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    45 packets output, 4748 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out
```

```
R2#sh ip int bri | include Tunnel0
Tunnel0      10.10.10.2      YES manual up      up
R2#
```

```
! Ruta LAN remota via Tunnel0 show
ip route
```

```

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.11.85.66 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.11.85.66
      10.0.0.0/8 is variably subnetted, 7 subnets, 3 masks
C      10.10.10.0/30 is directly connected, Tunnel0
L      10.10.10.1/32 is directly connected, Tunnel0
C      10.11.85.0/26 is directly connected, Ethernet0/1
L      10.11.85.1/32 is directly connected, Ethernet0/1
C      10.11.85.64/26 is directly connected, Ethernet0/0
L      10.11.85.65/32 is directly connected, Ethernet0/0
L      10.11.85.192/26 is directly connected, Tunnel0

```

! IKE Fase 1 show
crypto isakmp sa

```

R1#sh crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id status
10.11.85.130 10.11.85.65    QM_IDLE        1001 ACTIVE

IPv6 Crypto ISAKMP SA
R1#sh crypto isakmp sa

```

! IPsec Fase 2 – selectores proto 47 (GRE) show
crypto ipsec sa

```

R1#sh crypto ipsec sa
Interface: Ethernet0/0
  Crypto map tag: CM-GRE, local addr 10.11.85.65

protected vrf: (none)
local ident (addr/mask/prot/port): (10.11.85.65/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (10.11.85.130/255.255.255.255/47/0)
current_peer 10.11.85.130 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 45, #pkts encrypt: 45, #pkts digest: 45
  #pkts decaps: 45, #pkts decrypt: 45, #pkts verify: 45
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

local crypto endpt.: 10.11.85.65, remote crypto endpt.: 10.11.85.130
plaintext mtu 1458, path mtu 1500, ip mtu 1500, ip mtu idb Ethernet0/0
current outbound spi: 0xE83CBA83(3896294019)
PFS (Y/N): N, DH group: none

inbound esp sas:
  spi: 0xD2E468B9(3538184377)
    transform: esp-256-aes esp-sha256-hmac ,
    in use settings ={Transport,}

```

! ACL GRE con matches
show ip access-lists ACL-GRE-VPN

```

10 permit gre host 10.11.85.65 host 10.11.85.130 (47 matches)
R1#show ip access-lists ACL-GRE-VPN
Extended IP access list ACL-GRE-VPN
10 permit gre host 10.11.85.65 host 10.11.85.130 (57 matches)
R1#sh ip int brief | include tunnel0

```

! Crypto map

```
R1#sh crypto map
Crypto Map IPv4 "CM-GRE" 10 ipsec-isakmp
  Peer = 10.11.85.130
  Extended IP access list ACL-GRE-VPN
    access-list ACL-GRE-VPN permit gre host 10.11.85.65 host 10.11.85.130
  Current peer: 10.11.85.130
  Security association lifetime: 4608000 kilobytes/3600 seconds
  Responder-Only (Y/N): N
  PFS (Y/N): N
  Mixed-mode : Disabled
  Transform sets={
    TS-GRE: { esp-256-aes esp-sha256-hmac },
  }
  Interfaces using crypto map CM-GRE:
    Ethernet0/0
  Interfaces using crypto map NiStToSt1:
```

! Ping y Trace

```
PC1> ping 10.11.85.194
84 bytes from 10.11.85.194 icmp_seq=1 ttl=62 time=5.942 ms
84 bytes from 10.11.85.194 icmp_seq=2 ttl=62 time=2.081 ms
84 bytes from 10.11.85.194 icmp_seq=3 ttl=62 time=2.525 ms
84 bytes from 10.11.85.194 icmp_seq=4 ttl=62 time=1.181 ms
84 bytes from 10.11.85.194 icmp_seq=5 ttl=62 time=1.757 ms

PC1> trace 10.11.85.194
trace to 10.11.85.194, 8 hops max, press Ctrl+C to stop
 1 10.11.85.1 3.025 ms 0.577 ms 0.257 ms
 2 10.10.10.2 2.053 ms 1.254 ms 1.142 ms
 3 *10.11.85.194 1.328 ms (ICMP type:3, code:3, Destination port unreachable)

PC1> █
```

■ En 'show crypto ipsec sa' los selectores muestran protocol=47 (GRE), NO las IPs de las LANs. Esto confirma que IPSec protege el túnel GRE completo.